

Utah NASA Space Grant Consortium Fellowship Proposal

[Working Title — To Be Finalized]

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1 Introduction and Motivation

[TODO: Open with the problem statement and motivate why this research matters to NASA and the Utah NASA Space Grant Consortium’s space-related priorities. Briefly state the long-term vision and the specific gap this work addresses.]

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2 Background and Relation to NASA / Space Priorities

[TODO: Summarize the relevant scientific and engineering background, cite prior work, and explicitly tie the proposed research to NASA mission priorities (for example: in-space manufacturing, lightweight deployable structures, impact protection for crew or payloads, autonomous experimentation for space exploration, etc.). This subsection is currently a placeholder; the specific NASA alignment will be filled in in a later revision.]

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A representative early citation to verify the bibliography pipeline: [National Aeronautics and Space Administration, 2022, Skelton and de Oliveira, 2009].

3 Research Objectives

[TODO: State 2–3 concrete, measurable objectives for the fellowship year.]

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4 Proposed Approach and Methods

[TODO: Describe the methodology: experimental setup, computational tools, data-collection plan, and analysis pipeline. Highlight Marcus’s specific role and contributions.]

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Task Plan

[TODO: Replace placeholder tasks with the actual milestones.]

Task	Q1	Q2	Q3	Q4
Literature review and scoping	•			
Experimental / computational setup	•	•		
Data collection and analysis		•	•	
UNSGC conference presentation			•	
Manuscript / final report			•	•

Table 1: Quarterly task plan for the 2026–2027 fellowship year (placeholder).

5 Expected Outcomes and Broader Impact

[TODO: Describe deliverables (data sets, code, conference paper, manuscript, poster at the annual UNSGC conference) and the broader impact on NASA-relevant research, undergraduate training, and the Utah Space Grant mission.]

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6 Qualifications and Mentoring Plan

[TODO: Briefly describe Marcus's preparation (coursework, GPA, prior research or projects) and the advising structure with Prof. Baird, including weekly one-on-one meetings, lab group participation, and graduate co-mentoring.]

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References

National Aeronautics and Space Administration. NASA strategic plan. <https://www.nasa.gov/strategic-plan/>, 2022. Placeholder reference; replace with the most relevant NASA strategy or research-priority document for the final narrative.

Robert E. Skelton and Mauricio C. de Oliveira. *Tensegrity Systems*. Springer, New York, 2009. doi: 10.1007/978-0-387-74242-7.